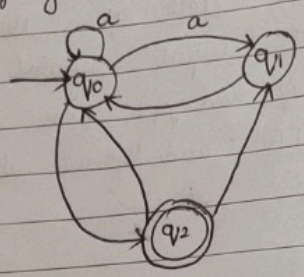
14/10/26

CO1 AT3

1. Consider the NFA given below which accepts some language L over the alphabet $\Sigma = \{a, b\}$. Construct a DFA that represents the same language L.



Transitions, $\Sigma = \{a, b\}$

Sol: Subset construction (NFA to DFA Conversion)

$\delta(q_0, a) = \{q_0, q_1\}$, $\delta(q_0, b) = \{q_2\}$

$\delta(q_1, a) = \{q_0\}$, $\delta(q_1, b) = \{q_1\}$

$\delta(q_2, a) = \{q_0\}$, $\delta(q_2, b) = \{q_1\}$

DFA transition table :

DFA States	Input aaa	Input bbb
$\rightarrow \{q_0\}$	$\{q_0, q_1\}$	$[q_2]$ (Accepting)
$[q_0, q_1]$	$[q_0, q_1]$	$[q_1, q_2]$ (Accepting)
$*[q_2]$	$[q_0]$	$[q_1]$
$*[q_1, q_2]$	$[q_0]$	$[q_1]$
$[q_1]$	$[q_0]$	$[q_1]$

3. Design takes string (either NFA ... by

10

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3. Design a Non deterministic Finite Automata (NFA) that takes a binary string as input and accepts the string only if the last two digits are the same (either 00 or 11). Write the formal definition of the NFA. Also show that the string 01011 is accepted by NFA.

Formal definition of the NFA :

$M = (Q, \Sigma, \delta, q_0, F)$ where

$Q = \{q_0, q_1, q_2, q_3, q_4\}$

$\Sigma = \{0, 1\}$

$q_0 = q_0$

$F = \{q_2, q_4\}$ (Accepting states for 00 and 11 respectively).

Transition function δ :

State	Input 000	Input 111
→ q_0	$\{q_0, q_1\}$	$\{q_0, q_3\}$
q_1	$\{q_2\}$	$\emptyset$
* q_2	$\emptyset$	$\emptyset$
q_3	$\emptyset$	$\{q_4\}$
* q_4	$\emptyset$	$\emptyset$

Trace for string 01011 :

- Start state : $\{q_0\}$
- Read 0 : $\delta(q_0, 0) = \{q_0, q_1\}$

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- Read 1 : $\delta(\{q_0, q_1, q_3\}, 1) = \{q_0, q_2\}$
- Read ϕ : $\delta(\{q_0, q_2, q_3\}, 0) = \{q_0, q_1\}$
- Read 1 : $\delta(\{q_0, q_1, q_3\}, 1) = \{q_0, q_3\}$
- Read 1 : $\delta(\{q_0, q_3\}, 1) = \{q_0, q_3, q_4\}$